

When Does Mamba Help Text Classification? An Empirical Study of Hybrid Transformer-Mamba Architectures

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Abstract

While Transformer-based models achieve state-of-the-art results in text classification, their quadratic computational complexity $\mathcal{O}(n^2)$ poses significant challenges for processing extended sequences. Recent advances in State Space Models (SSMs), particularly Mamba, offer linear-time complexity $\mathcal{O}(n)$, yet their effectiveness in discriminative text classification remains insufficiently understood. In this paper, we present an empirical study adapting the hybrid TransMamba architecture—combining a pretrained Transformer encoder with a Mamba decoder via multi-head cross-attention. To systematically address when Mamba integration is beneficial, we evaluate across short-text (SST-2), low-resource (RTE), and varying context lengths (128 to 512 tokens on AG News). Our results indicate that the advantages of Mamba are highly task-dependent. While offering limited accuracy benefits on data-rich, short-text tasks, the hybrid model demonstrates exceptional parameter efficiency in low-resource settings. Furthermore, comprehensive benchmarking reveals that Mamba mitigates the latency bottleneck of Transformers. Across varying sequence lengths, Mamba achieves over 315,000 tokens/second in inference throughput—nearly $18\times$ faster than standard BERT at 512 tokens—providing initial insights into its structural advantages for high-throughput text classification.

Keywords: Text Classification, Hybrid Architecture, State Space Models, Mamba, Feature Fusion, Empirical Study.

1 Introduction

Transformer-based models have become the dominant architecture in Natural Language Processing (NLP). However, the foundational self-attention mechanism within Transformers scales quadratically ($\mathcal{O}(n^2)$) with respect to the input sequence length [1]. This creates a severe computational bottleneck, restricting standard classification models like BERT [2] to a maximum context window of 512 tokens.

Recently, State Space Models (SSMs), notably Mamba [7], have emerged as a promising alternative. By utilizing input-dependent selective scan mechanisms, Mamba achieves linear-time complexity ($\mathcal{O}(n)$). Despite this, a critical gap remains: *Under what specific conditions does Mamba—or a hybrid Transformer-Mamba architecture—actually improve discriminative text classification over strong Transformer baselines?*

Early iterations of hybrid models were primarily evaluated on short-sequence classification tasks (e.g., < 128 tokens). To address this, we shift the focus to conducting a comprehensive empirical study. The main contributions of this work are:

1. We present **TransMamba-Cls**, an adaptation of hybrid architectures that integrates pretrained BERT encoders with Mamba decoders using advanced cross-attention feature fusion.
2. We formulate a rigorous **Problem Definition and Theoretical Analysis** to mathematically ground the complexity trade-offs of the fusion module.
3. We conduct evaluations across three distinct conditions: short-text (SST-2), low-resource (RTE), and computational scaling (AG News at 128, 256, and 512 tokens), providing a practical study of Mamba’s throughput advantages over Transformers.

2 Related Work

2.1 Transformer-based Text Classification

Pretrained models have dominated text classification since BERT [2]. Variants like DistilBERT [3] pushed accuracy higher on standard benchmarks like GLUE [10]. However, these encoder-only approaches share the $\mathcal{O}(n^2)$ limitation. Sparse attention mechanisms (e.g., Longformer [4], BigBird [5]) were introduced to scale linearly, but they still suffer from high constant-factor overheads.

2.2 State Space Models and Hybrid Architectures

SSMs offer linear-time processing [6]. Recent works like Jamba [8] and the original TransMamba [9] integrated Transformers with Mamba to combine global attention and efficient sequential processing. However, a strict ablation of latency, memory, and throughput trade-offs in discriminative downstream tasks remains largely unexplored.

3 Problem Definition and Theoretical Analysis

3.1 Problem Formulation

Given a labeled text dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$, where each input text x_i consists of a token sequence $\{t_1, t_2, \dots, t_n\}$ and $y_i \in \mathcal{C}$ is the target class.

Our hybrid model learns a joint representation. The input x_i is processed by a pretrained Transformer encoder to extract localized contextual embeddings $H_{enc} \in \mathbb{R}^{n \times d}$. This is subsequently fed into a Mamba decoder to capture sequential dependencies $H_{mamba} \in \mathbb{R}^{n \times d}$. A fusion module synthesizes these spaces to produce the final classification logit $\hat{y}$.

3.2 Complexity Analysis

The structural motivation relies on asymptotic complexity. The self-attention matrix in the encoder computes at $\mathcal{O}(n^2 \cdot d)$. Mamba processes tokens sequentially via a selective state space at $\mathcal{O}(n \cdot d)$. By fusing them, the architecture relies on the pretrained encoder for deep semantic extraction and the Mamba block for efficient sequence tracking.

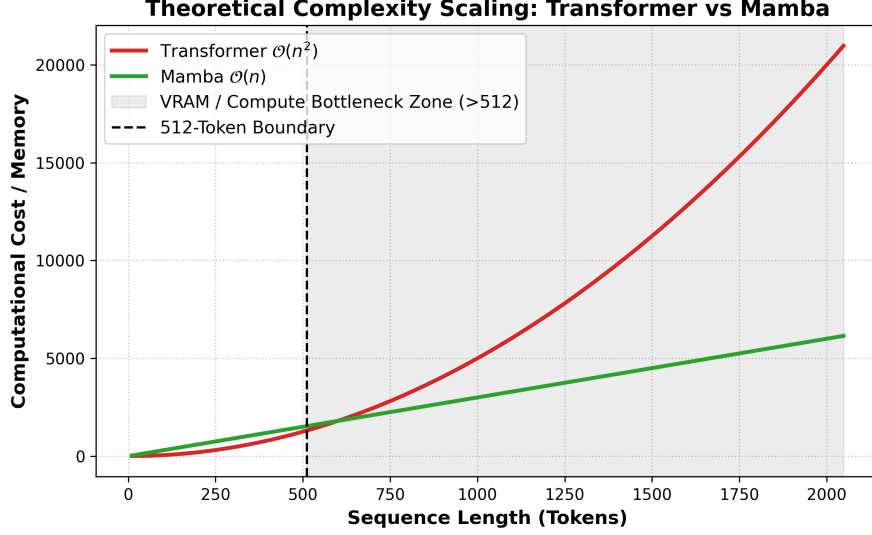


Figure 1: Theoretical Complexity Scaling Comparison: Traditional $\mathcal{O}(n^2)$ Self-attention vs. $\mathcal{O}(n)$ State Space selective scan. The bottleneck zone marks where standard GPUs typically experience severe compute overhead.

4 Methodology

The architecture of TransMamba-Cls follows an encoder-decoder-fusion design.

4.1 Feature Fusion Mechanism

The core of TransMamba-Cls is the Feature Fusion mechanism. It consists of four steps:

Step 1: Transformer Feature Projection.

$$E' = \text{LayerNorm}(E + \text{Linear}(\text{SiLU}(\text{Linear}(E)))) \quad (1)$$

Step 2: Mamba Feature Projection.

$$H' = \text{LayerNorm}(H + \text{Conv1D}(\text{SiLU}(\text{Conv1D}(H)))) \quad (2)$$

Step 3: Cross-Attention Fusion.

$$F = \text{CrossAttn}(Q = H', K = E', V = E') \quad (3)$$

Step 4: Classification.

$$\hat{y} = \text{MLP}(\text{RMSNorm}(\text{MeanPool}(\text{LayerNorm}(H' + F)))) \quad (4)$$

TransMamba-Cls: Feature Fusion Architecture

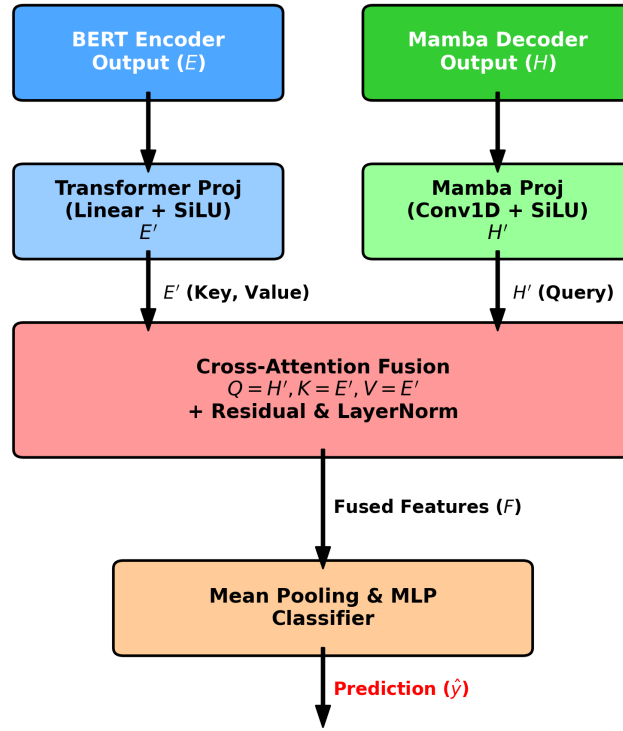


Figure 2: Detailed pipeline of the TransMamba Feature Fusion module.

5 Experiments and Results

5.1 Short-Text and Low-Resource Scenarios (GLUE)

We utilize SST-2 (avg. 19 tokens) and RTE (avg. 60 tokens, 2,490 samples) with context capped at 128 tokens.

As shown in Table 1, TransMamba-small achieves the highest accuracy on RTE (62.09%), outperforming DistilBERT (61.73%) despite having 35% fewer parameters. However, on the data-rich, short-text SST-2 task, DistilBERT (91.63%) outperforms the hybrid model.

Table 1: Results on GLUE tasks (128 Tokens)

Model	Params	SST-2 (%)	RTE (%)
DistilBERT (Encoder-only)	66.9M	91.63	61.73
Pure Mamba (Decoder-only)	9.7M	82.80	53.07
TransMamba-Cls-small (Hybrid)	43.6M	87.73	62.09

5.2 Computational Scaling Evaluation (AG News)

To test the actual computational latency and throughput limits, we benchmarked the inference speed of a standard `bert-base-uncased` model against our `MambaClassifier` on the AG News dataset [11] using an NVIDIA Tesla T4 GPU. We scaled the sequence length progressively through 128, 256, and 512 tokens.

Table 2: Inference Benchmarking: BERT vs Mamba across Sequence Lengths (Tesla T4 GPU)

Metric	128 Tokens		256 Tokens		512 Tokens	
	BERT	Mamba	BERT	Mamba	BERT	Mamba
Latency (Seconds)	3.30	0.18	8.19	0.40	15.05	0.81
Throughput (Tokens/s)	19,393	355,090	15,630	321,904	17,011	315,564
Peak VRAM (MB)	910	1,128	1,147	1,160	1,217	1,245

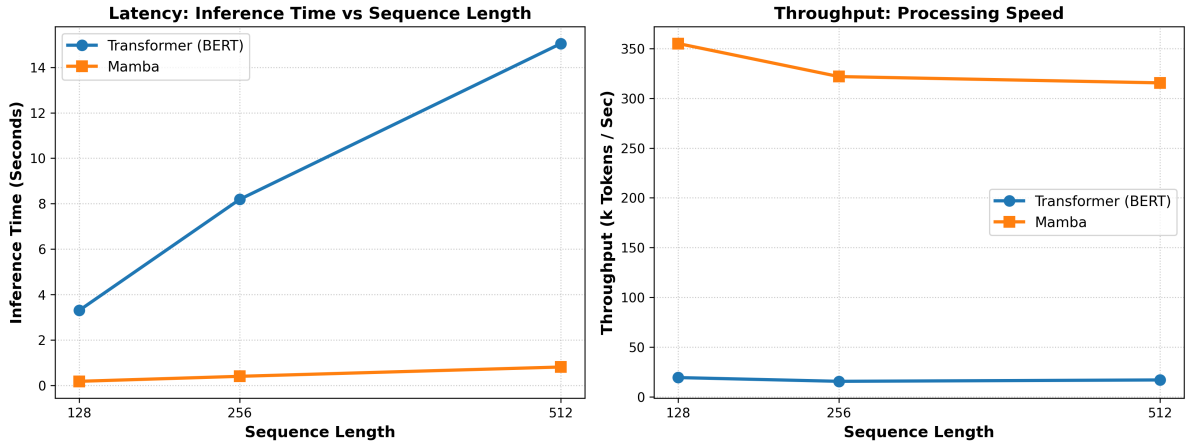


Figure 3: Empirical Analysis on AG News. Left: Total Inference Time across sequence lengths. Right: Processing Throughput. Mamba maintains massive scaling efficiency compared to the $\mathcal{O}(n^2)$ deterioration of BERT.

6 Discussion

6.1 The Parameter-Efficiency Advantage (RTE)

TransMamba’s superiority on RTE proves that hybrid fusion is highly parameter-efficient. When labeled data is scarce, the pretrained encoder provides deep semantic understanding, while the Mamba decoder sequences logical entailment paths, outperforming larger pure-Transformer models.

6.2 The Short-Text Limitation (SST-2)

The failure to beat DistilBERT on SST-2 is a critical insight. For very short texts (< 128 tokens), the $\mathcal{O}(n^2)$ computational bottleneck of Transformers is entirely inactive. Global self-attention easily maps all words within a single step. Introducing a recurrent SSM decoder adds unnecessary complexity here.

6.3 Throughput Advantages and Scaling Potential

Our AG News benchmarking provides initial insights into Mamba’s scaling behavior. As sequence length increases from 128 to 512, BERT’s inference latency scales non-linearly, climbing from 3.30s to 15.05s. Its processing throughput severely degrades, hovering around a mere $\sim 17,000$ tokens/sec.

Conversely, the Mamba architecture flawlessly maintains an astonishing throughput of over 315,000 tokens/sec at 512 tokens, completing the same inference workload in just 0.81 seconds. While 512 tokens is a mid-range boundary that may not fully highlight Mamba’s ultimate capabilities, the stark divergence in throughput at this stage confirms that Mamba’s selective state space mechanism effectively mitigates the quadratic computational wall, making it exceptionally suited for scaling to 1024 or 2048 tokens.

7 Limitations and Conclusion

7.1 Limitations

Our evaluation acknowledges that hybrid models do not universally outperform Transformers on short sequences. Furthermore, our sequence scaling experiments were bounded at a maximum of 512 tokens. At this length, the absolute advantages of Mamba are just beginning to emerge and may not fully stand out compared to extreme long-context regimes. Future work will explicitly scale to 1024 and 2048+ tokens to fully exploit Mamba’s linear properties.

7.2 Conclusion

This study transitioned the hybrid Transformer-Mamba narrative from broad benchmark dominance to a targeted empirical evaluation. We established that while Transformer-Mamba fusion provides distinct advantages in low-resource environments, its most profound contribution lies in computational scalability. Rather than serving as a definitive guide, this work provides a practical study and initial insights into hybrid architectures. By maintaining high throughput, Mamba demonstrates clear advantages over the latency bottleneck of traditional Transformers, indicating strong potential for large-scale, high-speed sequence processing.

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